

MPS

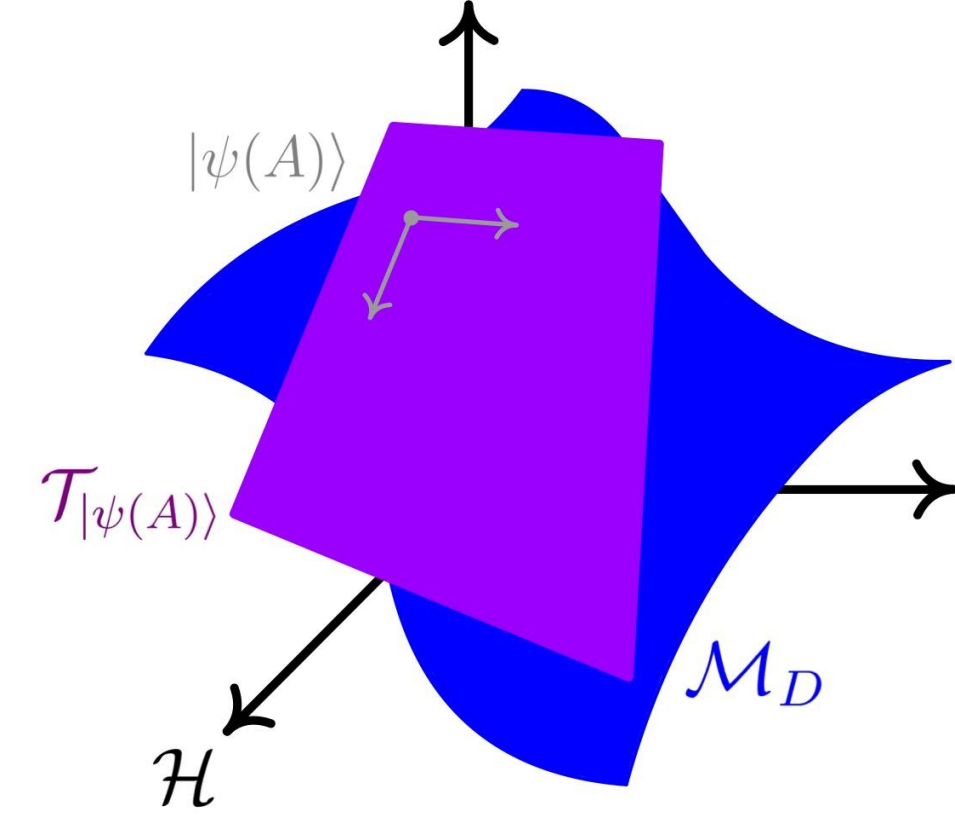
Manifold

$$|\psi(A)\rangle = [A^{[1]}] [A^{[2]}] \dots [A^{[N]}] \xrightarrow{\text{DMRG}} |\psi_0\rangle$$

Tangent space

$$|\psi(B; A)\rangle = \sum_n \sum_{s, \alpha, \beta} B_{\alpha, \beta}^{[n]s} \left[\partial_{A_{\alpha, \beta}^{[n]s}} |\psi(A)\rangle \right]$$

$$= \sum_n [A^{[1]}] \dots [A^{[n-1]}] B^{[n]} [A^{[n+1]}] \dots [A^{[N]}]$$



Quasiparticle excitation ansatz

Ground state $|\psi_0\rangle = [A_L^{[1]}] [A_L^{[2]}] \dots [A_L^{[N]}] C^{[N]} \leftarrow$

Tangent space $|\psi(B)\rangle = \sum_{n=1}^N [A_L^{[1]}] \dots [A_L^{[n-1]}] B^{[n]} [A_L^{[n+1]}] \dots [A_L^{[N]}] C^{[N]} \leftarrow$

Gauge freedom $B^{[n]} \rightarrow B^{[n]} + A_L^{[n]} Y^{[n]} - Y^{[n-1]} A_L^{[n]}$

Parametrization $B(X)$ such that (i) $\langle \psi_0 | \psi(X) \rangle = 0$, (ii) $X \mapsto |\psi(X)\rangle$ injective

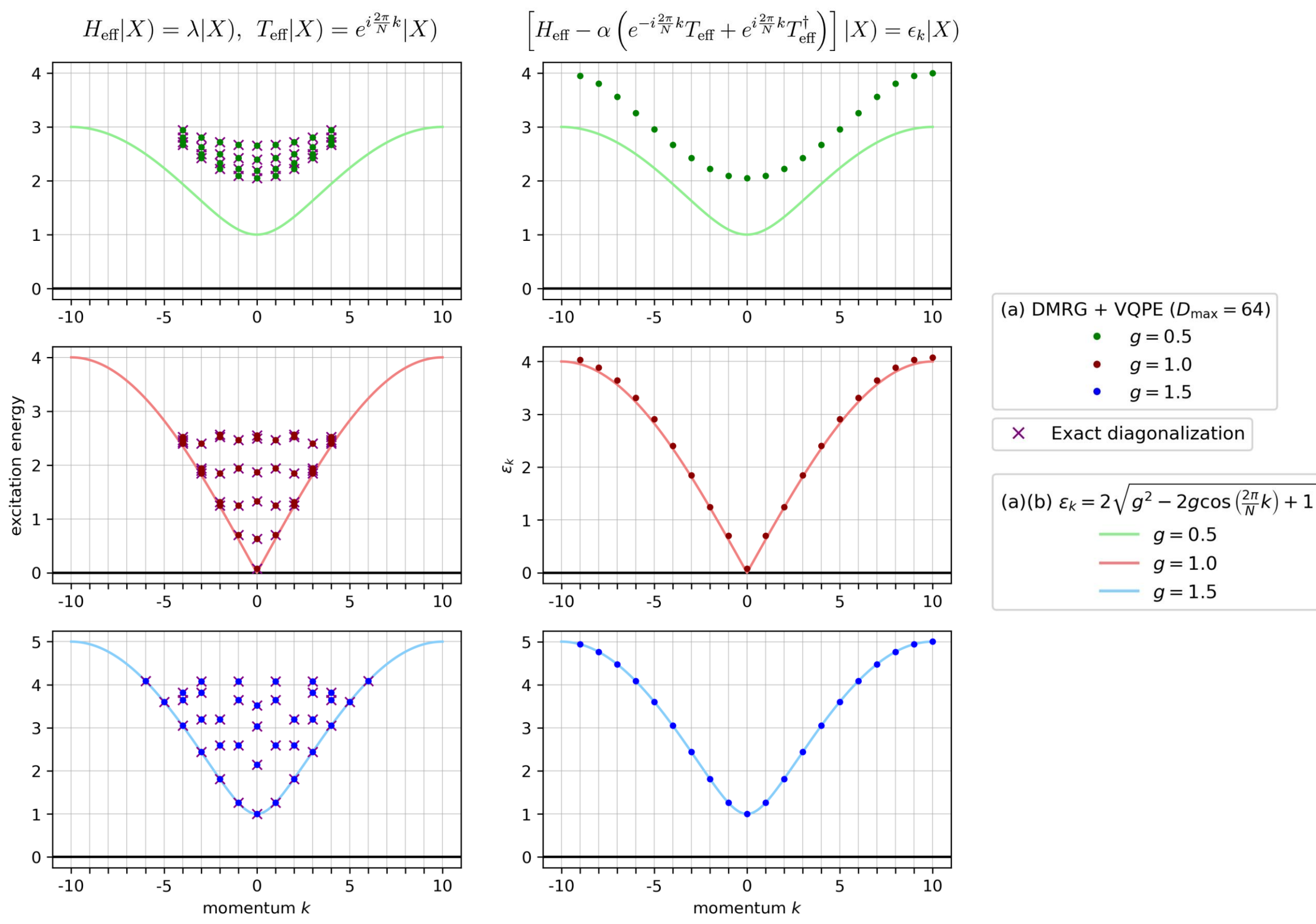
$$[B^{[n]}] = [V_L^{[n]}] [X^{[n]}] \quad \text{with} \quad \begin{cases} [V_L^{[n]}] \\ [X^{[n]}] \end{cases} = 0$$

$$|\psi(X)\rangle = \sum_{n=1}^N [A_L^{[1]}] \dots [A_L^{[n-1]}] [V_L^{[n]}] [X^{[n]}] [A_L^{[n+1]}] \dots [A_L^{[N]}] C^{[N]} \leftarrow$$

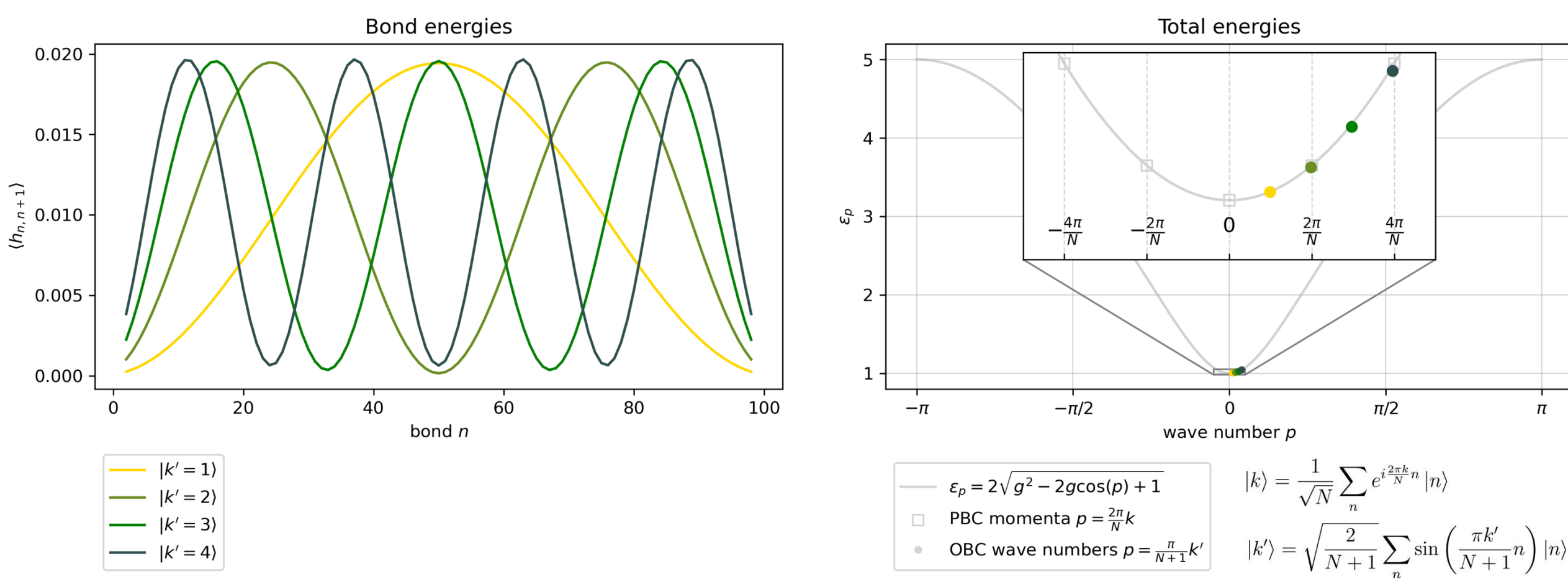
$$\Rightarrow \langle \psi_0 | \psi(X^{[n]}) \rangle = 0$$

$$\Rightarrow \langle \psi(\bar{Y}^{[m]}) | \psi(X^{[n]}) \rangle = \delta_{m,n} \langle \bar{Y}^{[m]} | X^{[n]} \rangle$$

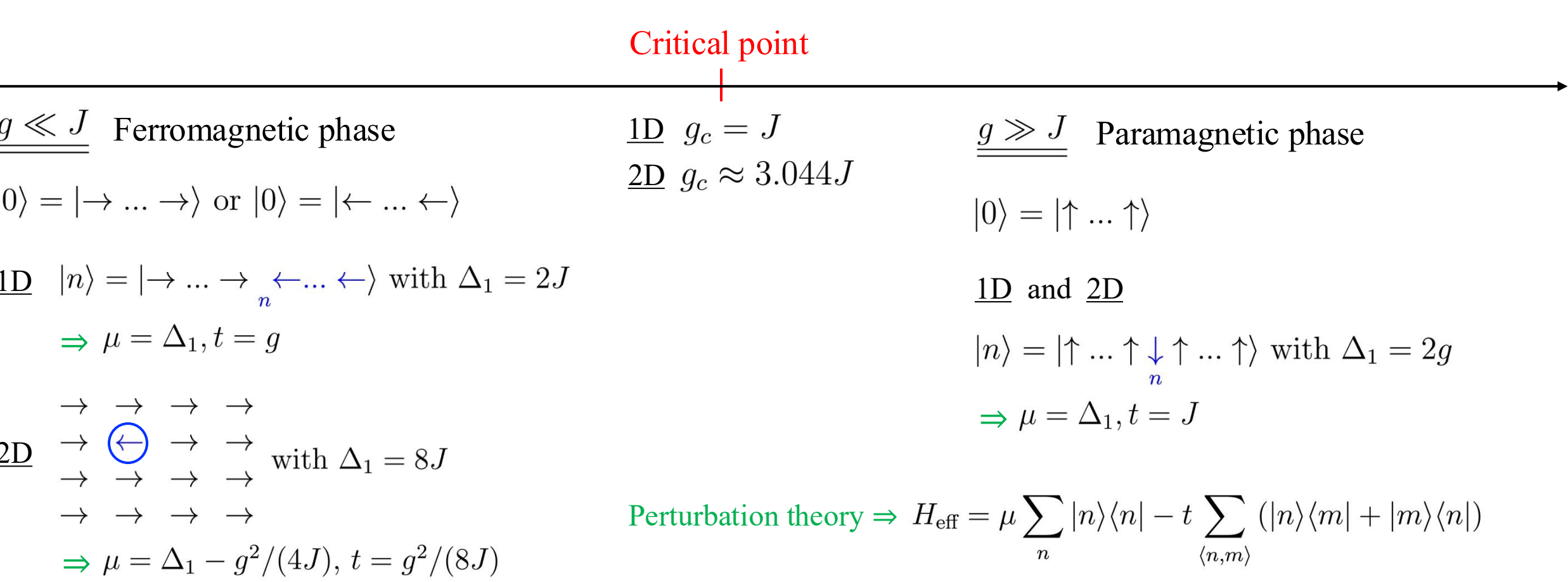
Restoring momentum with PBC Hamiltonian



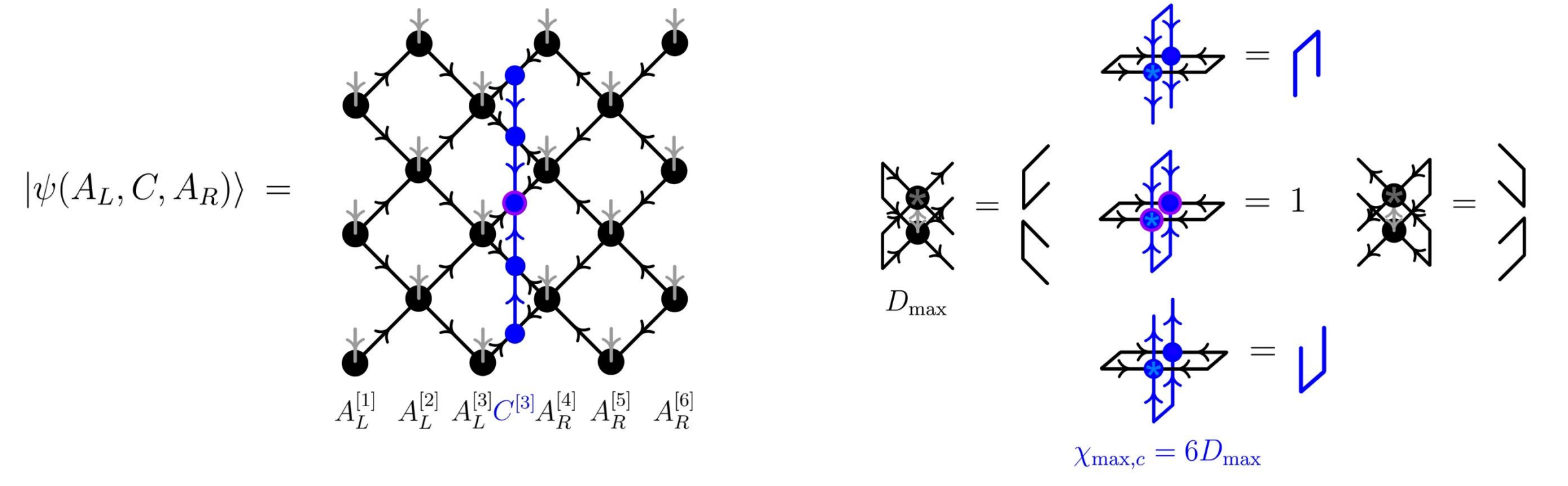
Standing waves with OBC Hamiltonian



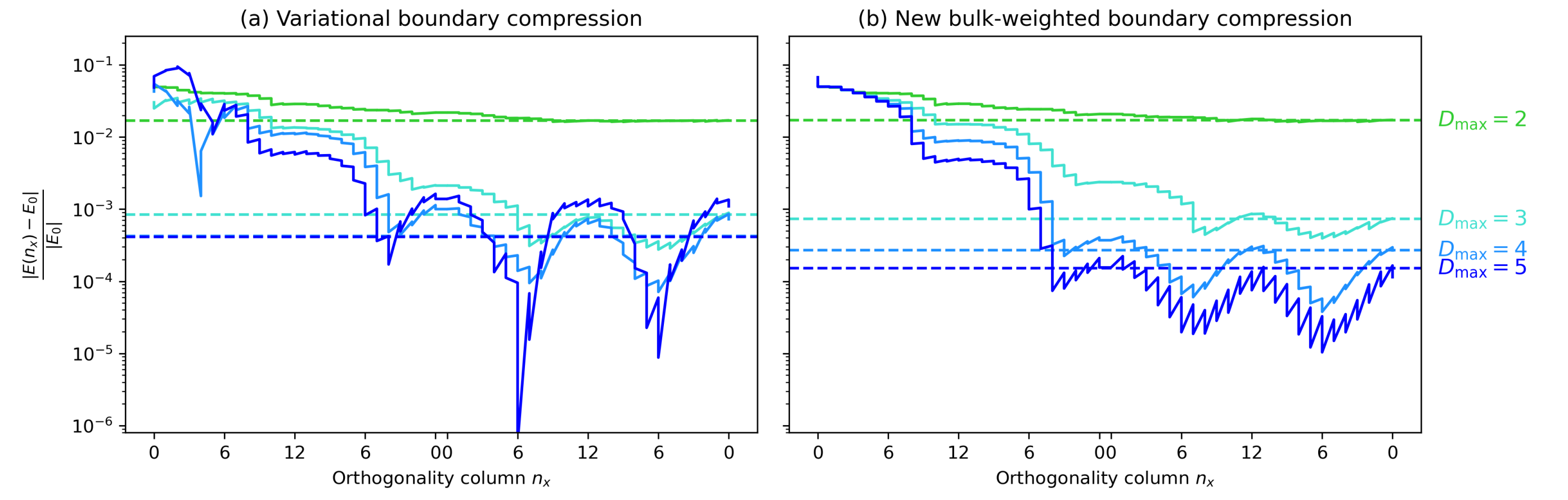
Transverse field Ising (TFI) model $H = -J \sum_{\langle n, m \rangle} \sigma_n^x \sigma_m^x - g \sum_n \sigma_n^z$



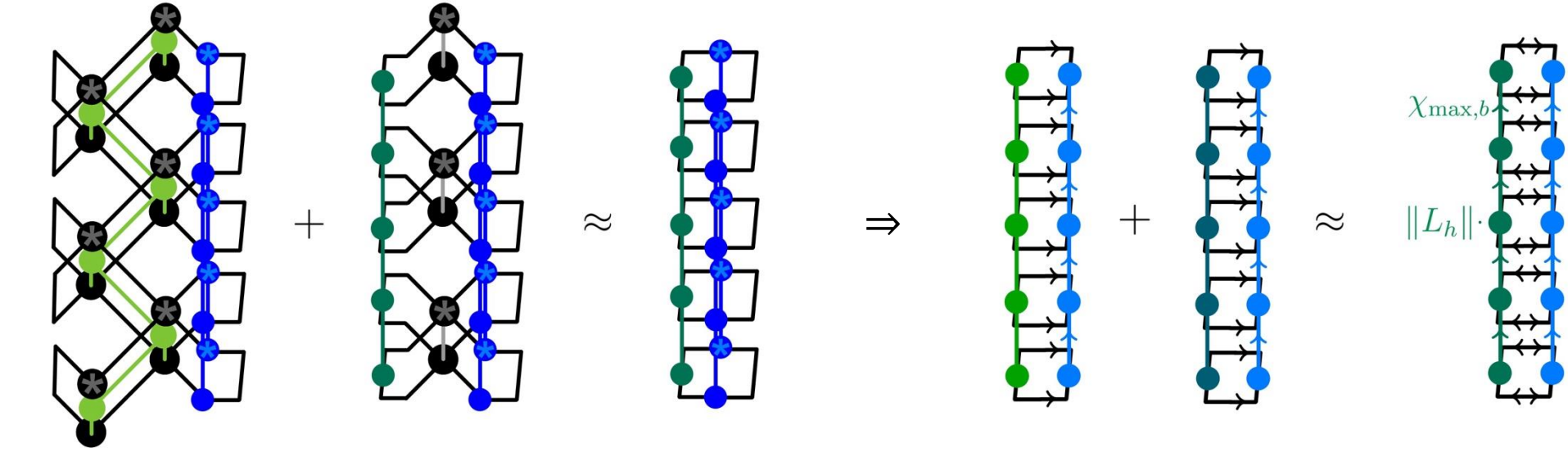
isoPEPS



Ground state with DMRG²



New bulk-weighted boundary compression

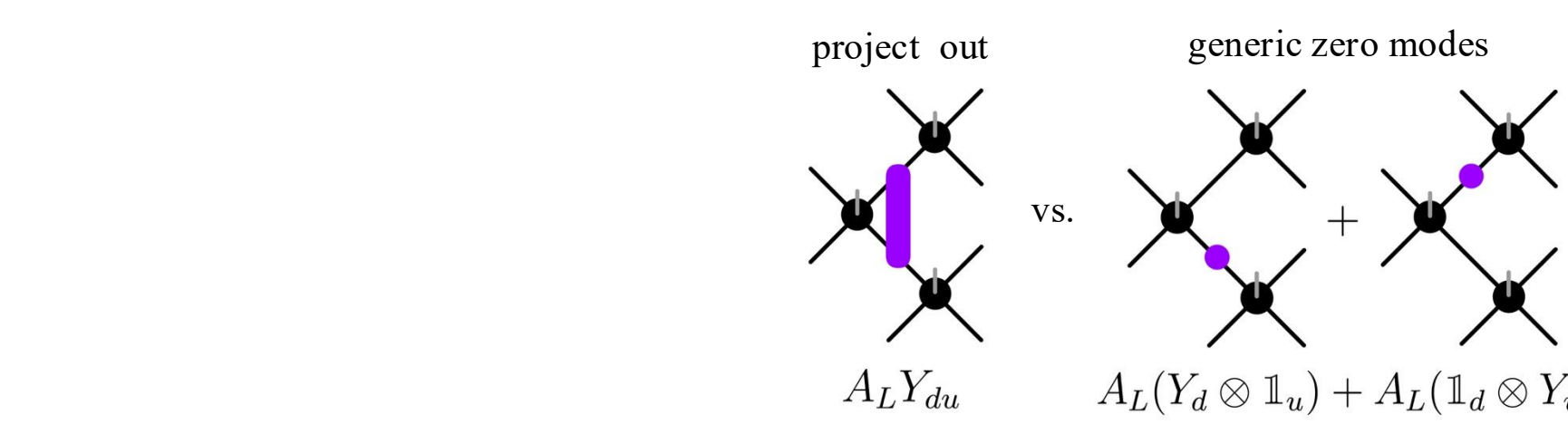
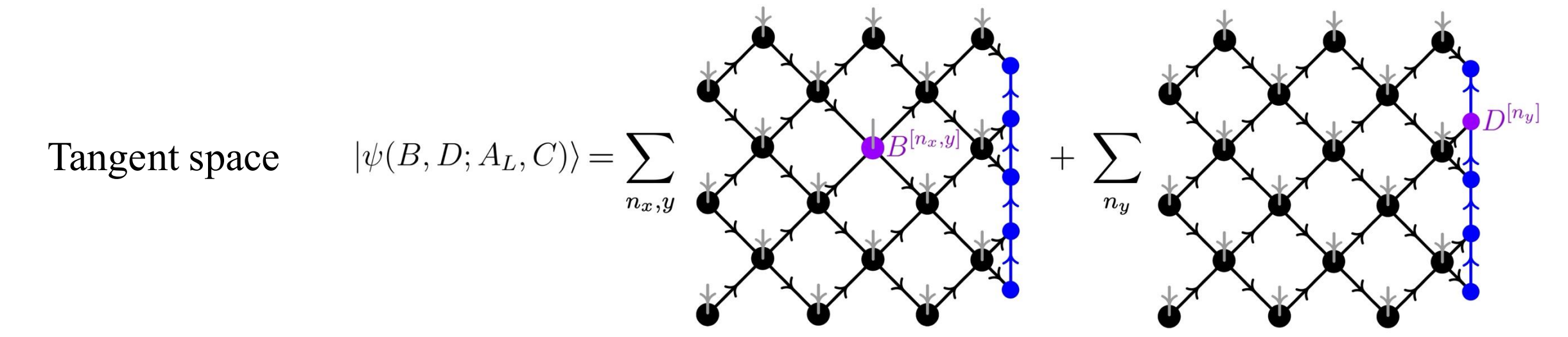


$$1) \quad L_h^{[1]} L_h^{[2]} \stackrel{\text{stacked QR}}{\approx} \begin{matrix} R_1 & R_2 \\ Q & Q \end{matrix} \quad (L_h^{[1]} L_h^{[2]}) = QR = Q(R_1 R_2)$$

$$\mathcal{O}(D_{\max}^3 \chi_{\max, b})$$

$$2) \quad \begin{matrix} L_h^{[1]} C \\ R_1 \end{matrix} + \begin{matrix} L_h^{[2]} C \\ R_2 \end{matrix} = L_h C \quad \xrightarrow{\text{SVD}} \begin{matrix} U & S & V \\ U' & & V' \end{matrix} = \begin{matrix} \tilde{L}_h^{[1]} R_1 \\ L_h \end{matrix} + \begin{matrix} \tilde{L}_h^{[2]} R_2 \\ L_h \end{matrix}$$

New quasiparticle excitation ansatz



Parametrization $B(X), D(X)$ such that (i) $\langle \psi_0 | \psi(X) \rangle = 0$, (ii) $X \mapsto |\psi(X)\rangle$ injective

$$B^{[n_x, y]} = V_L^{[n_x, y]} X^{[n_x, y]} \quad \text{with} \quad \begin{matrix} \bar{A}_L^{[n_x, y]} \\ V_L^{[n_x, y]} \end{matrix} = 0 \quad D^{[n_y]} = \begin{matrix} X^{[n_y]} \\ V_D^{[n_y]} \end{matrix} \quad \text{with} \quad \begin{matrix} \bar{C}_D^{[n_y]} \\ V_D^{[n_y]} \end{matrix} = 0$$

$$\Rightarrow \langle \psi_0 | \psi(X^{[n_x, y]}) \rangle = 0 \quad \Rightarrow \langle \psi_0 | \psi(X^{[n_y]}) \rangle = 0$$

$$\Rightarrow \langle \psi(\bar{X}^{[n_y]}) | \psi(X^{[n_x, y]}) \rangle = 0 \quad \Rightarrow \langle \psi(\bar{X}^{[n_y]}) | \psi(X^{[n_y]}) \rangle = \delta_{\bar{n}_y, n_y} \langle \bar{X}^{[n_y]} | X^{[n_y]} \rangle$$

$$\Rightarrow \langle \psi(\bar{X}^{[n_x, y]}) | \psi(X^{[n_x, y]}) \rangle = \delta_{\bar{n}_x, n_x} \delta_{\bar{y}, y} \langle \bar{X}^{[n_x, y]} | X^{[n_x, y]} \rangle$$

Diagonalizing the effective Hamiltonian

